# Reactions in plants

*9B Plant growth · Lesson 1 of 7*

Date _____ Class _____

## Today I can

- ☐ I can write the word and symbol equations for photosynthesis and aerobic respiration, and say where and when each happens in a plant.
- ☐ I can explain how limiting factors slow photosynthesis and why roots need oxygen.

## What happens today

1. Do Now: on a farm
2. Learn: Photosynthesis · Light, chlorophyll and limiting factors · Aerobic respiration · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 21 to 23 · Workbook pages 15 to 16

**What I learned today (one or two sentences)**

# Key Vocabulary

| Term | Definition |
|---|---|
| **photosynthesis** | the process plants use to make glucose from carbon dioxide and water, using light |
| **aerobic respiration** | the reaction in every living cell that releases energy from glucose using oxygen |
| **chlorophyll** | the green substance that captures energy from light |
| **chloroplast** | the part of a plant cell where photosynthesis happens |
| **glucose** | the sugar made in photosynthesis |
| **reactant** | a substance used up in a chemical reaction |
| **product** | a substance made in a chemical reaction |
| **limiting factor** | a variable that slows down the rate of photosynthesis |
| **phloem** | tubes made of living cells that carry sugar solution around a plant |
| **yield** | the amount of useful product a farmer gets from a crop |

# Quick Review

*on a farm*

Farmers want as much useful product (yield) from their crops as possible.

**1** What process do wheat plants use to make their own food?

**2** Name the gas that plants take in for this process.

**3** Suggest the effect of a very cloudy year on the wheat harvest.

# Core 1 of 3

*Everyone does these. Answer on the card or under it.*

**1** Complete the word equations. Use the word bank.

Photosynthesis: _____ + water makes _____ + oxygen

Aerobic respiration: glucose + _____ makes carbon dioxide + _____ (+ energy released)

**2** Tick one box in each row to show whether the statement is about photosynthesis or aerobic respiration.

| Statement | Photosynthesis | Respiration |
|---|---|---|
| happens only when there is light | | |
| happens all the time, day and night | | |
| takes in carbon dioxide | | |
| releases energy from glucose | | |
| happens in chloroplasts | | |
| happens in every living cell | | |

# Core 2 of 3

*Everyone does these. Answer on the card or under it.*

**3** Match each key word to its meaning. Draw a line.

| | |
|---|---|
| chlorophyll | tubes of living cells that carry sugar solution around the plant |
| chloroplast | the sugar made in photosynthesis |
| limiting factor | the green substance that captures energy from light |
| phloem | a variable that slows down the rate of photosynthesis |
| glucose | the part of a plant cell where photosynthesis happens |

**4** Van Helmont grew a willow tree in a pot of dry soil for five years. He added only rainwater. The table shows his results.

| | At the start | After five years |
|---|---|---|
| mass of dry soil (kg) | 90.72 | 90.66 |
| mass of tree (kg) | 2.27 | 76.74 |

**a)** Calculate the change in the mass of the soil.

# Core 3 of 3

*Everyone does these. Answer on the card or under it.*

**4** *continued*

**b)** Calculate the change in the mass of the tree.

**c)** Aristotle thought that roots sucked up soil for the plant to eat. Explain how these results show that Aristotle was wrong.

# Challenge

*When Core is finished. Full sentences.*

**C1** Draw a flow chart to show how energy from the Sun becomes stored in a plant. Use the words light, chlorophyll, chloroplast and glucose.

*Draw here.*

**C2** After his experiment, van Helmont concluded that all the extra mass of the tree came from the water he added. Evaluate this conclusion. Use the word equation for photosynthesis in your answer.

# Exit Ticket

*Hand this card in at the end of the lesson.*

Name _____

**1** Write the word equation for photosynthesis.

**2** Where in a plant cell does photosynthesis happen?

**3** Why do plants in a field only photosynthesise during the day?

**Today I can:** Goal 1 ☐ Goal 2 ☐